

Impaired Percent Alpha Variability on Continuous Electroencephalography Is Associated with Thalamic Injury and Predicts Poor Long-Term Outcome after Human Traumatic Brain Injury

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ABSTRACT

Continuous electroencephalography (cEEG) is potentially useful in determining prognosis in patients with traumatic brain injuries (TBI). The objective of this prospective, observational cohort study was to determine if the percent alpha variability (PAV) on cEEG was predictive of outcome following TBI. Injury characteristics were indexed to assess whether lesions in specific cerebral loci were correlated with PAV and patient recovery. Fifty-three TBI patients were studied using cEEG recording and serial neuroimaging. Clinical recovery was assessed at regular intervals in hospital and following discharge. The principal outcome measures included the mean 3-day PAV score, the 7-day PAV pattern, delineation of the anatomical sites of brain injury, and the 6-month clinical outcome, as measured by the Glasgow Outcome Scale (GOS). Significant univariate ($p = 0.030$) and multivariate ($p = 0.008$) relations were identified between PAV and GOS scores. PAV offered good discrimination between favorable and unfavorable 6-month outcomes (AUC 0.76) and, with a cut-point of 0.20, had a sensitivity of 87% and negative predictive value of 82%. Multivariate modeling revealed that injuries of the thalamus ($p = 0.009$) and basal ganglia ($p = 0.016$), and the presence of diffuse edema ($p = 0.009$), were the key anatomical predictors of PAV. Brainstem injuries ($p = 0.020$) and indicators of diffuse cerebral trauma, such as deep white matter shearing ($p = 0.036$) and multiple subcortical lesions ($p = 0.033$), were the principal determinants of 6-month recovery. Inclusion of PAV enhanced the accuracy of prediction models that encompassed a selective combination of clinical and anatomical variables (adjusted $R^2 = 0.458$, $p < 0.001$). The two main results of this study are (1) PAV is a sensitive predictor of 6-month clinical outcomes following TBI, and (2) injury to the thalamus is related to impaired PAV. PAV appears best utilized as a functional adjunct to traditional clinical and anatomical predictors.

Key words: adult brain injury; electrophysiology; outcome measures; recovery; secondary insult

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INTRODUCTION

TRAUMATIC BRAIN INJURIES (TBI) have a tremendous socioeconomic impact and are a major consumer of health care resources in North America (Brain Trauma Task Force, 2000; Kraus and McArthur, 1996; NIH Consensus Development Panel, 1999). The ability to assess patient prognosis early in the course of hospital treatment facilitates clinical decision-making and permits accurate family counseling to impart realistic expectations for recovery. Several clinical and radiographic measures (e.g., Glasgow Coma Scale [GCS] score, pupil reactivity, age, lesions on early computed tomography [CT] scans) have significant predictive value when studied alone or in combination; however, the heterogeneous structure of regional trauma systems produces variability in acute TBI care and the interpretation of baseline measures, thus limiting generalizability and confidence in many prediction models (Choi et al., 1991).

There is renewed and growing interest in the significance of the electrophysiological alterations that occur in the brains of TBI patients. With exception, the frequency content recorded by cortical electroencephalography (EEG) correlates with cerebral blood flow, metabolism and function (Baldy-Moulinier and Ingvar, 1968; Ingvar et al., 1976). Alpha waves predominate in the occipital regions, but may also be found over Rolandic and temporal cortices, and are marked by frequency (i.e., 8–13 Hz), reactivity to associated stimuli (e.g., light, audition, cognitive tasking), and variability of signal. These features typify EEG recordings during the relaxed, conscious state (Niedermeyer, 1997). The source and regulatory mechanisms that govern the alpha rhythm remain controversial; however, there is abundant animal data that implicate thalamic regions in its inception and control (Hughes and Crunelli, 2005). The evidence in humans is less robust, and there is little known about the neural mechanisms that alter the alpha rhythm following TBI in humans. We have previously described correlations between the relative or percent alpha variability (PAV) and the clinical outcomes of patients with aneurysmal subarachnoid hemorrhage (SAH) or TBI. Alpha variability was significantly reduced prior to, and during, symptomatic ischemia in patients with SAH-induced vasospasm. These changes signaled the need for urgent intervention and were typically reversed with resolution of ischemia (Nuwer, 1994; Vespa et al. 1997). Similar monotonous trends were found in TBI patients with altered brain glucose metabolism and persistent coma (Bergsneider et al., 1997). It was subsequently demonstrated that reduced PAV, measured over the initial 3 days in hospital, provided early and reliable prediction of poor 30-day outcomes in patients with moderate to severe TBI. Further-

more, incorporation of PAV markedly improved the accuracy of prediction models comprised of traditional variables alone (Vespa et al., 2002).

While several population studies have been performed outlining predictive factors, the current goal was to rigorously study a small group of patients to determine if specific deficits in the PAV or the anatomic substrates of PAV are influential in determining outcome after TBI. We hypothesized that a reduction in PAV would forecast a poor long-term (6-month) outcome and correlate with injuries to the thalamus and/or adjacent deep brain structures in this population. Continuous EEG (cEEG) monitoring and serial neuroimaging was performed in a novel cohort of TBI patients to determine if early PAV measures, recorded over the first 3–7 days after injury, were reliable predictors of 6-month clinical outcome. Cerebral injury characteristics were systematically catalogued to reveal the anatomical correlates of PAV and provide novel evidence of a putative alpha rhythm generator in the human brain.

METHODS

Subjects, Imaging, and Clinical Assessments

This study was approved by the University of California at Los Angeles (UCLA) Institutional Review Board (IRB). Patients admitted to the neurosurgical intensive care unit (NICU) were prospectively enrolled if the following inclusion criteria were met: (1) direct admission to the UCLA Medical Center with a diagnosis of TBI, (2) a GCS score of 13 or lower, (3) age 16–85 years, and (4) an abnormal head CT. Patients were excluded if the initial neurological exam was consistent with brain death or if the GCS improved to 14 or greater within 8 h of admission. Patients who received pentobarbital- or propofol-induced burst suppression to manage intracerebral hypertension or respiratory failure were also excluded because burst suppression is associated with a reduction in PAV. Other exclusion criteria included pre-existing neurological disorders and overwhelming concurrent hepatic or metabolic encephalopathy.

As part of the study protocol, patients underwent serial radiographic imaging of the brain during the hospital course and throughout the year following injury. The imaging studies were interpreted by one of six neuroradiologists who were blinded to the details of the study. Two neuroradiologists read 91% of the representative CT and magnetic resonance imaging (MRI) scans that were included in subsequent analyses (139 of 153 scans). Injury characteristics were documented from radiology reports by a blinded observer. Regions of interest were the basal ganglia (i.e., caudate, putamen, globus pallidus),

thalamus, brainstem, and deep white matter (i.e., corona radiata, internal capsule, cerebral peduncle, corpus callosum). The presence of diffuse edema (i.e., cerebral swelling that spanned more than one cortical lobe or subcortical structure) and multiple sites of subcortical injury were also evaluated.

The neurological status of patients was monitored hourly while in the ICU and at least every 4 h during the remainder of the hospital stay. Patients were followed in clinic after discharge and outcomes were scored at the 6-month assessment using the Glasgow Outcome Scale (GOS: 1, dead; 2, vegetative state; 3, severe disability; 4, moderate disability; 5, good recovery).

Electroencephalography Monitoring Protocol

Continuous EEG monitoring was performed in the NICU, as previously described (Vespa et al., 1999). The International 10-20 System was used to configure a 10-electrode montage that incorporated eight channels of CZ-referenced activity and eight channels of bipolar derivations. The EEG data were then further processed by fast-Fourier transform to generate spectra of percent alpha (PA) activity. The bandwidth for the PA used in this study was 6–14 Hz (expanded alpha), which has been standardized in previous reports and shown to correlate with cerebral blood flow and metabolic rate of oxygen (Ingvar et al., 1976; Vespa et al., 1999, 2002). Monitoring was conducted for 7 days and interrupted only for transportation of the patient for radiological imaging, diagnostic testing, or procedures performed outside of the NICU. The EEG was continuously displayed at the bedside and available for the attending nurses and physicians to review. Specific EEG characteristics (e.g., denoting the presence of seizures) were identified by the treating team and managed appropriately. All EEG data were stored digitally onto a computer hard drive or data tape for archival purposes.

Treatment Protocol

Patients who required an initial emergency operation had postoperative cEEG monitoring following transfer to the NICU. Intracranial pressure (ICP) monitoring was performed using ventriculostomy and/or pressure transducer systems, and the ICP was kept below 20 mm Hg with a stepwise management strategy using cerebrospinal fluid drainage, administration of mannitol, and mild hyperventilation to PaCO₂ of 30–35 mm Hg. Cerebral perfusion pressure was kept equal to or greater than 60 mm Hg by using phenylephrine, when required. Patients with a GCS score of eight or less, with normal coagulation parameters, underwent jugular venous oximetry monitoring using a no. 4 French oximetry catheter. All patients re-

ceived a phenytoin loading dose (14–18 mg/kg) on admission to the emergency room and continued on phenytoin maintenance for at least 7 days. Daily total serum phenytoin levels were determined, and additional boluses were given to maintain serum levels at 10–20 mg/dL. Sedation was maintained using either intermittent intravenous midazolam and morphine or continuous infusions of these drugs. The EEG segments recorded during sedation with greater than 2 mg/h of midazolam or morphine, or during infusion of propofol at doses higher than 20 micrograms per kilogram per minute, were excluded from this study as drug effects on cEEG and PAV can be significant at these levels (Vespa et al., 1999).

Scoring of the cEEG Trend

Scoring of PA trends was modified from the methods outlined in previous reports (Vespa et al., 1997; 2002). Histograms of PA were printed at an identical scale (15% per seven millimeter height) every 8 h and later scored for variability (i.e., PAV) by a physician who was blinded to the clinical scenario and patient outcome. The raw EEG waveforms were reviewed to identify artifacts that may have contaminated the trends. All trends that contained electrical artifact or were recorded during periods of excessive sedation or (post-) ictus were excluded from scoring. The PAV score was determined by visually identifying three points on the PA histogram: the PA baseline, the PA peak value, and the PA trough value that most directly follows the peak. The PA baseline is the mean PA occurring during the 4 h prior to a peak in PA. The PAV is calculated using the following formula:

$$(\text{Peak PA} - \text{Trough PA}) \div (\text{Peak PA} + \text{Trough PA})$$

This yields a fraction value with a range of 0.05 to 0.50 (Vespa et al., 2002). EEG data were recorded in epochs of 4–12 h. The cEEG epoch that contained activity patterns observed in at least 50% of the daily trace obtained for a designated electrode pair was considered representative for that day and used for PAV scoring. Four PAV measures were calculated per day using bilateral frontotemporal (F4-T4) and temporoparietal (T4-P4) bipolar electrode derivations. These derivations were chosen to reflect the global convexity activity. The global daily PAV scores were calculated as the mean of the four hemispheric PAV values. The 3-day PAV scores were calculated by averaging the individual daily values that were available between day 0 (admission day) and hospital day 3.

Statistical Analysis

The primary outcome measures of this study were (1) mean 3-day PAV scores, (2) 7-day PAV patterns, (3) cat-

egorically indexed brain injury profiles, and (4) 6-month GOS scores. Spearman correlation analysis was performed to evaluate the relationship between the 3-day PAV and 6-month GOS scores. Two sample *t*-tests or Mann-Whitney Rank Sum tests and chi-square tests of independence were used to investigate relationships between groups of ordinal and categorical data, respectively. One-way analysis of variance (ANOVA) was used to evaluate differences between more than two groups of data. Logistic regression modeling of 3-day PAV and 6-month GOS scores was performed using the anatomical sites of injury identified by neuroimaging (Littell et al., 1996). Model simplification progressed by selective elimination from initial, overparameterized models that incorporated all possible predictors, to a final, optimized combination of variables. Fit was evaluated using residual and quantile-quantile plots. The ability of PAV to discriminate between unfavorable (GOS 1–3) and favorable (GOS 4–5) 6-month clinical outcomes was assessed using receiver operating characteristic (ROC) curve analysis calculated with package ROCR in R v2.11 (R Development Core Team, 2005; Sing et al., 2004). Unless

otherwise specified, data are expressed as mean $\pm$ standard deviation (SD).

RESULTS

Patient Demographics and Percent Alpha Variability

This study prospectively enrolled 53 patients; their demographics are outlined in Table 1. At the UCLA Medical Center, patients with TBI and a GCS score of 13 or less were admitted to the NICU. The median and mean GCS scores at presentation were 7.0 and 7.64 ± 3.1 , respectively. Many patients exhibited clinical deterioration during the initial assessment period, and the median and mean GCS scores prior to transfer to the NICU were 7.0 and 6.51 ± 1.84 , respectively. The cohort had a mean age of 41.5 ± 21.2 years. Fourteen patients were excluded from the PAV analysis because of subsequent sedation requirements in the NICU. Two patients were lost to follow-up and did not have GOS scores calculated at the 6-month assessment. Sixty-four percent (25 of 39) of pa-

TABLE 1. PATIENT DEMOGRAPHICS

<i>Demographic</i>	<i>Category</i>	<i>Number of patients (%)</i>
Age, years	≤ 30	24 (45)
	31–60	18 (34)
	61–85	11 (21)
Gender	Male	40 (75)
	Female	13 (25)
Cause of TBI	Fall	20 (38)
	Vehicular crash	17 (32)
	Pedestrian struck by vehicle	10 (19)
	Gunshot wound	4 (7)
	Blunt trauma	2 (4)
Presenting GCS score	≤ 8	40 (76)
	9–12	6 (11)
	13–15	7 (13)
GCS score on admission to NICU	≤ 8	50 (94)
	9–12	2 (4)
	13–15	1 (2)
Neuroimaging	CT	53 (100)
	MRI	39 (74)
Surgical decompression	Yes	29 (55)
	No	24 (45)
Outcome measure obtained	PAV	39 (74)
	6-month GOS	51 (96)
	Lesion index	53 (100)
	PAV + 6-month GOS	38 (72)
	PAV + lesion index	39 (74)
	6-month GOS + lesion index	51 (96)

TBI, traumatic brain injury; GCS, Glasgow Coma Scale; NICU, neonatal intensive care unit; CT, computed tomography; MRI, magnetic resonance imaging; PAV, percent alpha variability; GOS, Glasgow Outcome Scale.

tients with acceptable PAV data had cEEG initiated within 24 h of NICU admission; 95% (37 of 39) of these patients had cEEG initiated within 48 h. The total duration of recording varied amongst patients and ranged from 3 days to 3 weeks. A total of 412 daily alpha spectra were obtained during the initial week in the NICU, and the 7-day PAV trend for each patient could be categorized into one of the following patterns, as previously described: I, progressive improving (10 patients); II, stable high (13 patients); III, stable low (11 patients); IV, progressive deteriorating (five patients; Fig. 1) (Vespa et al., 2002). The attainment of a single PAV measure above the cutpoint of 0.20 was used to differentiate between patterns II and III.

Relations between PAV Scores, PAV Patterns, and Clinical Outcomes

There was a significant positive correlation between the 3-day PAV and 6-month GOS scores ($r = 0.47$; $p = 0.003$; Fig. 2A). The PAV scores varied markedly amongst different GOS categories ($p = 0.044$), with sig-

nificantly lower average values in those who had an unfavorable (GOS 1–3, 0.15 ± 0.015) versus favorable (GOS 4–5, 0.20 ± 0.007) recovery ($p = 0.009$; Fig. 2B). The cohort of patients who did not achieve a minimum mean PAV score of 0.15 had a significantly lower mean GOS score at 6 months (2.69 ± 1.54), compared with the group whose mean PAV score was 0.15 or greater (4.18 ± 1.26 ; $p = 0.002$; Fig. 2C).

The mean 3-day PAV score varied significantly amongst patient groups that exhibited different patterns of PAV over the first week in hospital ($p < 0.0001$) with a marked reduction in the mean score associated with patterns III and IV (0.14 ± 0.04) compared with patterns I and II (0.22 ± 0.08 ; $p = 0.0003$; Fig. 3A,B). There was a significant tabular relation between the measure of a PAV score greater than 0.15 and the presence of PAV patterns I or II, compared with patterns III or IV ($\chi^2 = 15.4$, $p < 0.0001$). Corresponding differences were observed amongst the GOS scores in groups that exhibited different PAV patterns (Fig. 3C; $p = 0.003$). Patterns I and II were associated with significantly better 6-month outcomes (GOS score 4.23 ± 1.23) compared to those

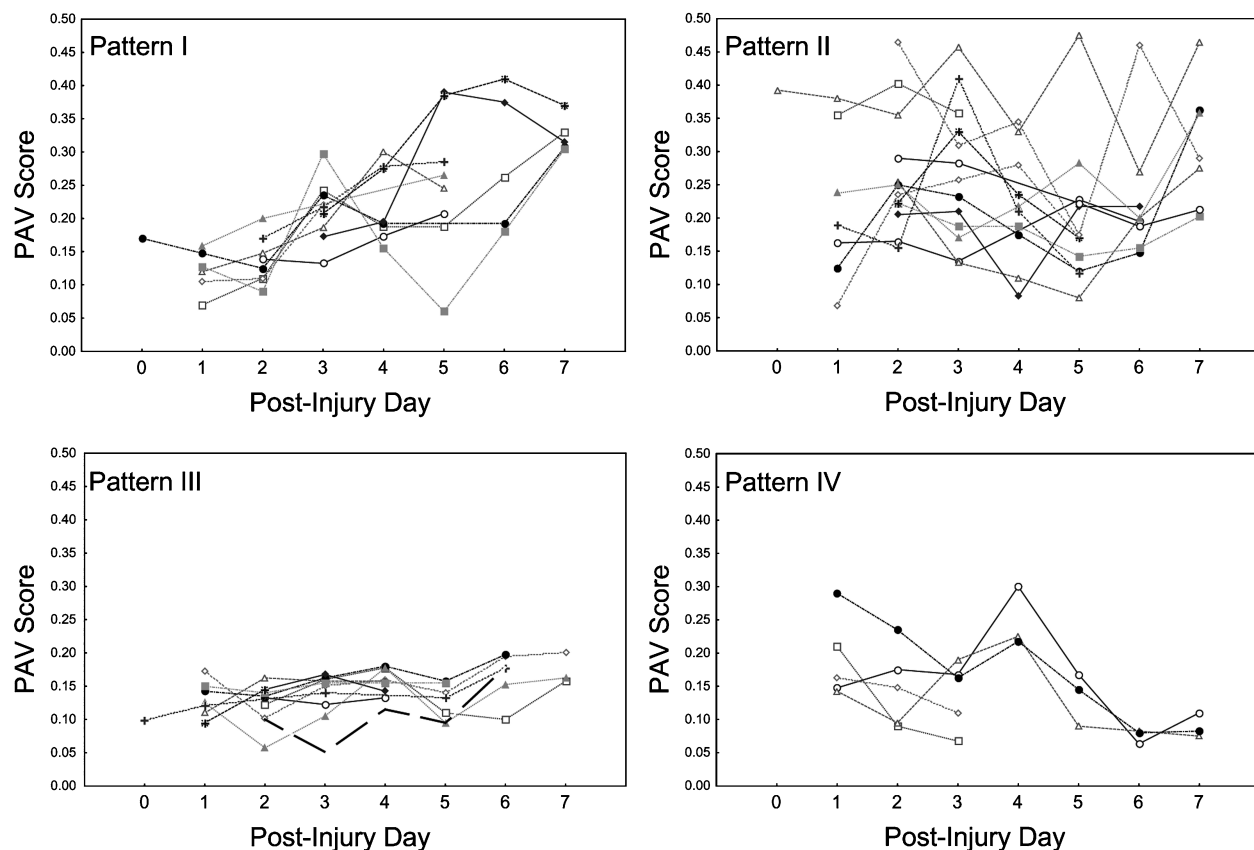


FIG. 1. Temporal characteristics of percent alpha variability (PAV). Four PAV patterns were exhibited over the first 7 days in neonatal intensive care unit (NICU): I, progressive improving; II, stable high; III, stable low; IV, progressive deteriorating.

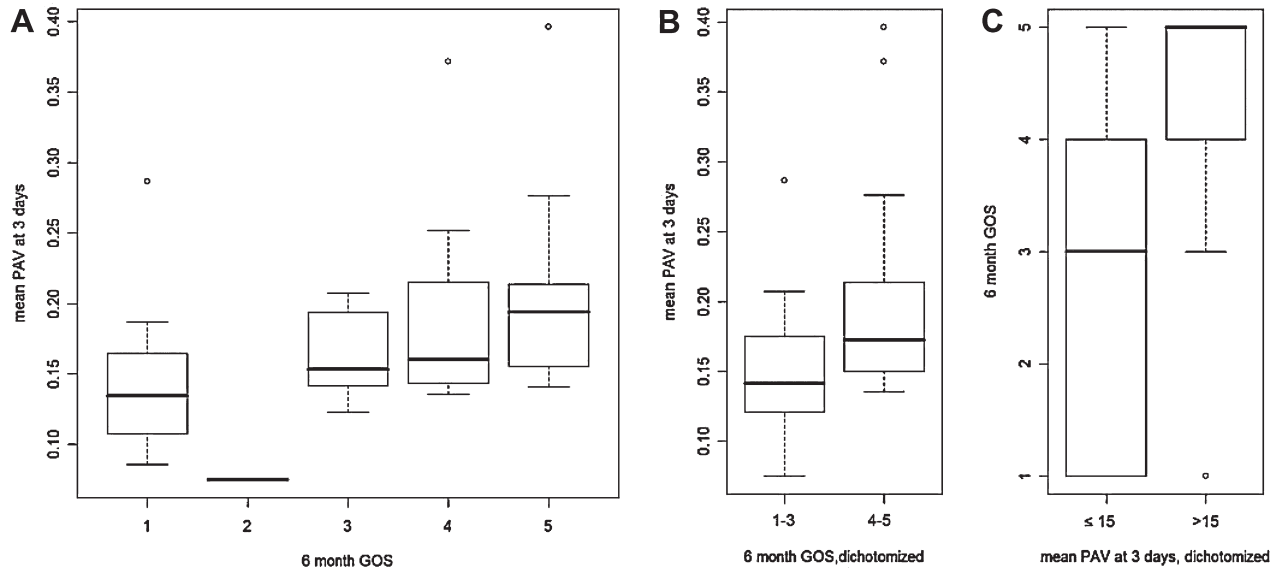


FIG. 2. Relations between percent alpha variability (PAV) and Glasgow Outcome Scale (GOS). (A) There was a significant, positive correlation between the mean 3-day PAV and 6-month GOS scores ($r = 0.47$; $p = 0.003$). (B) Patients who progressed to a favorable recovery (GOS 4–5) had significantly greater mean 3-day PAV scores compared with those who fared poorly (GOS 1–3; $p = 0.009$). (C) Conversely, there was a significantly greater incidence of unfavorable recovery (GOS 1–3) in patients who attained a mean 3-day PAV score of 0.15 or less, when compared to those with higher PAV scores ($p = 0.004$). Boxplots show the median scores (bold line), upper and lower quartiles (box), and minimum and maximum values (whiskers) in each group. Data outliers are indicated by open circles.

who exhibited patterns III or IV (GOS score 2.63 ± 1.50 ; $p = 0.0009$; Fig. 3D). Chi-square testing revealed a consistent, reciprocal relationship between the PAV pattern and the GOS score (i.e., patterns I and II were associated with higher GOS scores; $\chi^2 = 7.91$, $p = 0.005$).

Anatomical Correlates and Predictive Modeling of the PAV

PAV data were dichotomized at values of 0.15 (i.e., the mean PAV score of the unfavorable outcome group) and 0.20 (i.e., the mean PAV score of the favorable outcome group) and evaluated for lesion characteristics and frequency. No significant relations were identified using the lower cutpoint; however, with the PAV dichotomized at 0.20, the presence of multiple subcortical lesions was significantly associated with lesser PAV scores ($\chi^2 = 4.8$, $p = 0.029$). Multivariate stepwise regression using traditional variables of sex, age, pupil abnormalities, and initial GCS score offered poor prediction of PAV, with no single variable achieving significance (adjusted $R^2 = 0.042$, $p = 0.247$). Prediction was markedly improved by evaluating the anatomical sites of injury (adjusted $R^2 = 0.224$, $p = 0.025$), with thalamic lesions being the key predictor of PAV ($p = 0.008$). Clinical and anatomical markers were combined and selectively eliminated to cre-

ate a reduced model that included thalamic injuries ($p = 0.009$), diffuse edema ($p = 0.009$), basal ganglia injuries ($p = 0.016$), and age ($p = 0.040$), with further improvement of PAV prediction (adjusted $R^2 = 0.283$, $p = 0.004$).

Anatomical Correlates and Predictive Modeling of Six-Month Recovery

Patients were dichotomized into groups of unfavorable (GOS 1–3) and favorable (GOS 4–5) recovery and evaluated for the frequency of cerebral lesions, previously described. Significant positive relations were identified between poor outcome at 6 months post-injury and the presence of diffuse edema ($\chi^2 = 22.1$, $p < 0.0001$), multiple subcortical lesions ($\chi^2 = 9.2$, $p = 0.004$), or brainstem injuries ($\chi^2 = 6.5$, $p = 0.020$). Regression modeling revealed that the combination of sex ($p = 0.692$), age ($p = 0.001$), pupil abnormalities ($p = 0.044$), and presenting GCS score ($p = 0.022$) offered significant prediction of 6-month GOS scores in this cohort (adjusted $R^2 = 0.268$, $p < 0.001$). The anatomical markers alone produced moderately better results, with diffuse edema ($p = 0.010$) and the presence of multiple lesion sites ($p = 0.039$) being the key predictors of 6-month GOS scores (adjusted $R^2 = 0.314$, $p < 0.001$). A selective model that

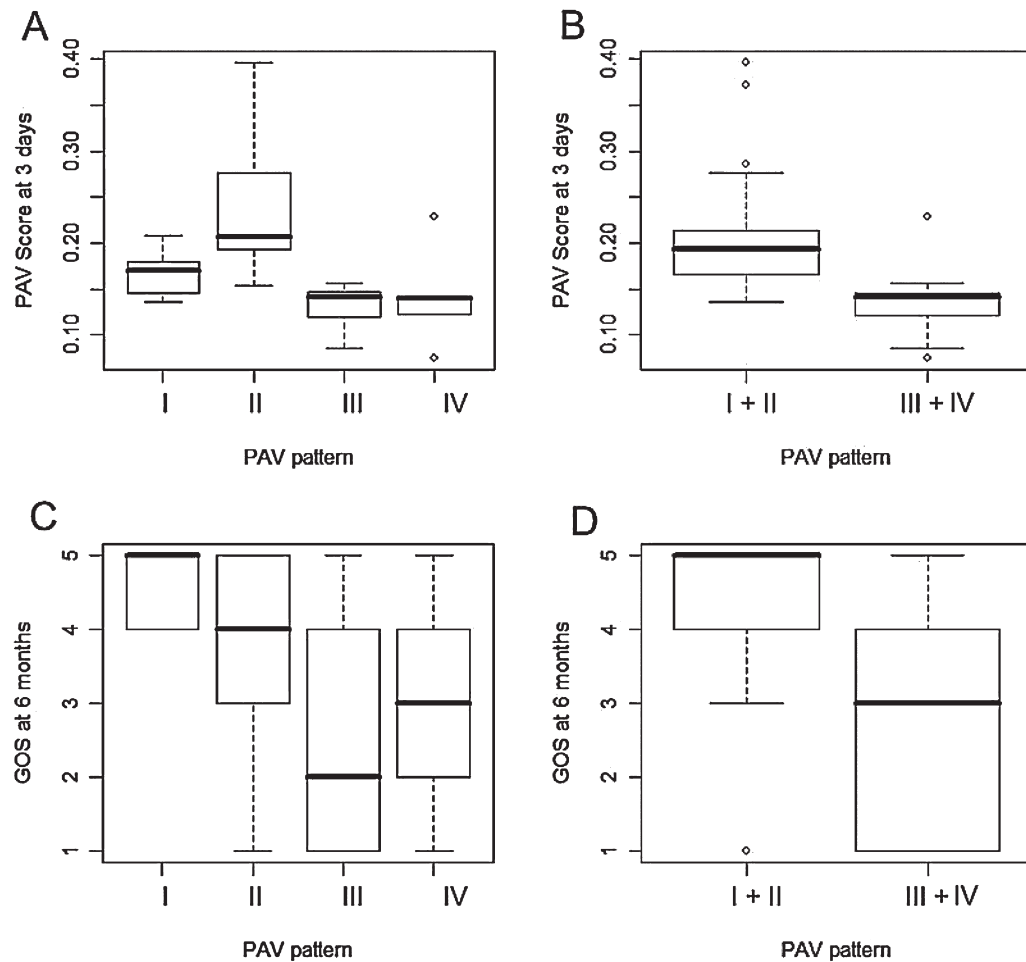


FIG. 3. Percent alpha variability (PAV) pattern correlations. Significant variation was observed in PAV ($p < 0.0001$) (A) and GOS ($p = 0.003$) (C) scores amongst the four PAV pattern groups. There were significantly higher mean PAV scores ($p < 0.0001$) (B) and GOS scores ($p = 0.002$) (D) in patients who exhibited patterns I or II, compared with III or IV. Boxplots show the median values (bold line), the upper and lower quartiles (box), and the minimum and maximum values (whiskers) in each group. Data outliers are indicated by open circles.

incorporated the anatomical markers, with white matter ($p = 0.036$) and multiple lesions ($p = 0.033$) significant, as well as pupil abnormalities ($p = 0.027$) and 3-day PAV scores ($p = 0.008$) offered further improvement over models that used exclusive combinations of clinical or anatomical variables (adjusted $R^2 = 0.458$, $p < 0.001$).

Predictive Accuracy and Receiver Operating Characteristic Analysis of PAV

Patient outcomes were dichotomized at the 6-month interval as unfavorable (i.e., dead, vegetative, or severely disabled) or favorable (i.e., moderately disabled or good recovery) and analyzed as a function of the mean 3-day

PAV score. The area under the receiver operating characteristic (ROC) curve was 0.76 (Fig. 4). Using a cut-point of 0.20, PAV analysis had a sensitivity of 0.87 and negative predictive value of 0.82. These values indicate that 87% of patients who progress to an unfavorable outcome (i.e., disease positive) will have had a PAV score less than 0.20 (i.e., test positive) and that 82% of patients who achieve a PAV score greater than or equal to 0.20 (i.e., test negative) will progress to a favorable recovery (i.e., disease negative). Predictive accuracy varied systematically across a range of PAV thresholds, with low scores associated with greater specificity and positive predictive values, and higher scores associated with greater sensitivity and negative predictive values (Table 2).

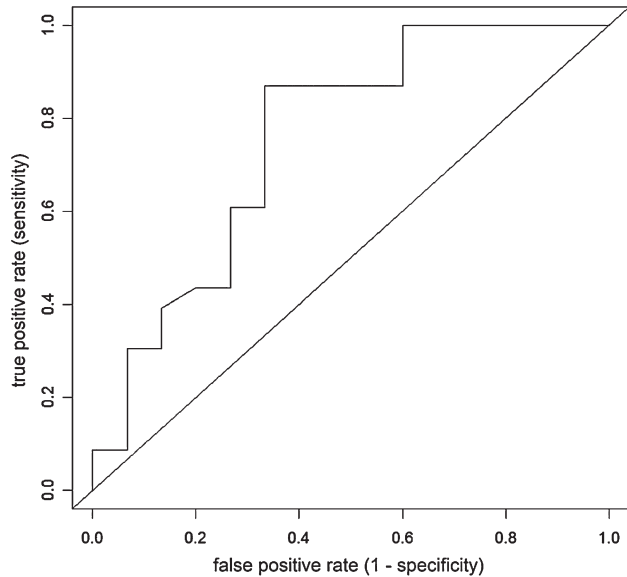


FIG. 4. Receiver operating characteristic (ROC) curve analysis. Area under the curve was 0.76.

DISCUSSION

The most common EEG changes associated with severe brain injury are frequency slowing and predominance of delta and theta bands. However, alpha oscillations can persist in neurologically compromised patients in a state of “alpha coma,” and two distinct anomalous alpha patterns have been described (Iragui and McCutchen, 1983; Westmoreland et al., 1975). Global cerebral dysfunction produced by TBI or metabolic encephalopathy may be associated with a pan-cortical alpha signal that is non-reactive to somatosensory stimulation. In contrast, focal brainstem lesions may produce reactive, posterior-predominant signals that mimic those in the awake, healthy brain. Alpha coma has typically been considered an ominous clinical sign, and the reduction of

alpha variability may be the key predictor of poor outcomes in these patients (Kaplan et al., 1999). We have previously shown that reduced PAV is a sensitive and reliable predictor of an unfavorable, 30-day outcome following TBI (Vespa et al., 2002). The current study has extended these findings by investigating the relations between early hospital PAV measures, the anatomical sites of brain injury and the long-term (six-month) clinical outcomes in a novel cohort of TBI patients.

Early PAV Measures Predict Six-Month Outcomes following TBI

The current data demonstrate a positive association between the mean 3-day PAV scores or the 7-day PAV patterns, and the 6-month clinical outcomes in this patient cohort. Those who progressed to a favorable recovery generally had PAV scores greater than 0.15, with a group mean PAV score of 0.20. We have previously examined the relations between PAV and 30-day outcomes following TBI. In that study, persistent PAV scores of 0.1 or lower, or worsening of PAV scores to 0.1 or lower were associated with a high likelihood of a vegetative recovery or death (Vespa et al., 2002). The current results complement those data and provide novel insight into the utility of PAV in making predictions of long-term outcome. Extrapolation from both datasets suggests that, while low PAV scores are a harbinger of an unfavorable outcome and high scores predictive of a favorable outcome, there is an intermediate range (i.e., PAV 0.10–0.20) within which the predictive accuracy of these measures may wane. If the PAV score falls within this interval, the PAV pattern may further assist in prognostication, as patients with persistently low scores or deteriorating trends (i.e., patterns III or IV) have been shown to fare poorly in both early and late outcome studies. In addition, PAV scores can significantly enhance the accuracy of more conventional prediction models. Rather than selecting a fixed PAV threshold upon which to base

TABLE 2. ACCURACY OF PAV

PAV cutpoint	Specificity (%)	Sensitivity (%)	Positive predictive value (%)	Negative predictive value (%)
0.10	100	20	100	66
0.15	74	66	63	77
0.20	39	87	48	82
0.25	17	93	42	80

Incremental threshold values of PAV were assessed for predicting favorable or unfavorable outcomes. Mean 3-day PAV scores at or below 0.1 were highly predictive of poor recovery or death, whereas scores of 0.20 or higher supported a favorable outcome. PAV scores of 0.1–0.2 had lower predictive accuracy.

PAV, percent alpha variability.

clinical interpretation, it is valuable to recognize and utilize the trade-off between specificity and sensitivity, and positive and negative predictive values, at incremental PAV cutpoints.

Thalamic Lesions Are Key Predictors in Anatomical Models of PAV

The thalamus has come to be recognized as a central regulator of the alpha rhythm. Many of the contemporary ideas about alpha physiology have been shaped by experimental animal studies. Early work that investigated the relations between neocortical and thalamic regulation of drug-induced, sleep spindles showed that, while cortical oscillations required intact thalamocortical circuitry, there persisted thalamic spindle activity in decorticate subjects (Morrison and Bassett, 1968). It was subsequently demonstrated that spindle events were driven by phasic thalamic inhibitory (i.e., GABAergic) interneurons that produced post-inhibitory, alpha-frequency, firing of thalamocortical cells. This led to the suggestion that cortical rhythms may be under the enduring influence of multiple thalamic pacemakers (Andersen and Andersson, 1968; Steriade et al., 1997; von Krosigk et al., 1993). However, sleep spindles are akin to, but clearly distinct from, true alpha activity and the relations between the two electrical phenomena remain unclear. Recordings in dogs and cats identified true alpha activity and demonstrated its distinct character from sleep spindles in these species. There was a strong correlation between the alpha activity in the visual thalamus and its associated neocortex, with results of coherence analyses supportive of a role for both thalamocortical and intracortical circuitries in the generation and regulation of alpha activity (Lopes da Silva et al., 1973; 1980).

In humans, thalamic injuries have been reported to variably influence alpha activity (Ohmoto et al 1978; Yazawa et al., 2001), and studies that have used EEG, combined with functional MRI or positron emission tomography, have presented contrasting results (Danos et al., 2001; Goldman et al., 2002; Larson et al 1998; Lindgren et al 1999). We have previously reported correlations between PAV flattening and altered level of consciousness in patients with SAH and vasospasm-induced ischemia of the thalamus or basal ganglia (Vespa et al., 1997). Consistently, pharmacological manipulation of EEG frequency content (e.g., with lorazepam) has since been used to demonstrate a high correlation between the state of arousal, lateral thalamic metabolism, and cortical alpha power in healthy human subjects (Schreckenberger et al., 2004). The current findings further support a fundamental role of the thalamus in the generation or maintenance of alpha variability in patients with TBI. All

combinations of anatomical predictors that were tested demonstrated a significant influence of thalamic lesions in the determination of PAV. The contribution of basal ganglia injuries, multiple subcortical lesions and diffuse edema was also significant and may have produced alpha changes directly or through involvement of nearby or en passant thalamic projections or regional modulatory nuclei.

Brainstem and Diffuse Brain Injuries Predict Poor Outcome after TBI

Age, presenting GCS score, and pupillary abnormalities are independent predictors of recovery after TBI (Brain Trauma Task Force, 2000). The influence of gender on patient outcome has not been well established and conclusions from available studies range from no gender effect (Combes et al., 1996; Klauber et al., 1989; Slew-Younan et al., 2004; Tooth et al., 2001; White et al., 2001) to either male (Grosswater et al., 1998; Signorini et al., 1999; Thornhill et al., 2000) or female (Farace and Alves, 2000; Kraus et al., 2000; Morrison et al., 2004) vulnerability to a worse prognosis. Taken together, these findings are consistent with the results of the traditional model presented here, in that age, GCS score, and pupillary abnormalities, but not gender, were significant predictors of clinical outcomes at 6 months following TBI. Marshall et al. (1991) described a CT classification of TBI based on the patency of the basal cisterns, the degree of midline shift, and the presence of an intracranial hematoma. This classification scheme predicted the development of intracranial hypertension or a fatal outcome and correlated with 6-month GOS scores (Lobato et al., 1997; Stocchetti et al., 2000; Vos et al., 2001). CT findings of compressed or obliterated basal cisterns, extensive traumatic subarachnoid hemorrhage, midline shift of more than 1.5 cm, the presence of a mass lesion, or severe diffuse cerebral injury were independently associated with a positive predictive value greater than 70% for poor outcomes (Brain Trauma Task Force, 2000). Several groups have demonstrated the independent association of brain swelling and multiple contusions with poor functional outcomes or patient mortality (Fearnside et al., 1993; Liu et al., 1995; Lobato et al., 1983; Marshall et al., 1991). Magnetic resonance (MR) imaging permits greater delineation of lesion characteristics and has broadened the evidence that brainstem injuries are an ominous sign that adds significant prognostic value to radiography-based prediction models (Firsching et al., 2001, 2002; Wedekind et al., 2002). Consistently, we have shown that indicators of diffuse brain injury on radiological studies (e.g., deep white matter shearing, diffuse edema, multiple contusions), as well as the presence

of brainstem injuries, are key predictors of an unfavorable outcome at 6 months post-ictus. Not unexpectedly, predictive accuracy was superior in models that incorporated both clinical variables and lesion characteristics, and was optimized with further consideration of the mean 3-day PAV score.

Study Limitations

The necessary use of neuroleptic medications in specific cases may have influenced the EEG data, and all patients under deep sedation were excluded from analysis. The study population was, therefore, limited to individuals with moderate to severe TBI who were treated with the standard of neurocritical care, including medical and surgical management of intracranial hypertension, with subsequent low sedation requirements. Other potentially confounding variables were patient age and co-morbidities. PAV was highly correlated with both thalamic lesions and 6-month clinical outcomes; however, a specific relation between thalamic lesions and the extent of recovery was not demonstrated by the current data. The reason for this finding remains speculative but likely reflects the relatively small patient numbers. Larger studies that permit subgroup analysis will flush out the importance of these issues and facilitate validation of the current results. The mechanistic link between PAV and cerebral function remains an enigma that also requires further investigation.

CONCLUSION

The mean 3-day PAV score and 7-day PAV pattern provided reliable estimates of 6-month patient outcomes following TBI. The current findings are consistent with the notion that the thalamus and adjacent subcortical structures are key anatomical determinants of PAV. In this study, outcome prediction was optimized in models that utilized PAV and a select combination of clinical and anatomical variables. PAV measures may significantly enhance the predictive accuracy of traditional models and, in the cases where baseline clinical variables are ambiguous, can serve in combination with radiographic findings or as a stand-alone means of early outcome prediction for TBI patients.

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